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D20A/59

M3/13

Demonstrate how the nonce prevents repeated processing of the same Ethereum transaction from an externally owned account.

Ethereum is an **account-based** blockchain. An **Externally Owned Account (EOA)** is controlled by a private key and can only act by sending signed transactions. Because a signed transaction is just data, anyone who sees it could try to submit it again (a **replay attack**). Ethereum defends against this using the **nonce**.

What is a Nonce?

The nonce is a per-account counter equal to the number of transactions already sent from that EOA. Every transaction carries a nonce field, which is part of the data that the sender signs, and the network increases the account's nonce by 1 once a transaction is included in a block. Transactions from one account are therefore numbered sequentially and processed strictly in that order.

Rule and Formula

$$\textit{Transaction } T \textit{ is valid} \iff T.\textit{nonce} = N(\textit{account})$$

$$\textit{After execution: } N(\textit{account}) \leftarrow N(\textit{account}) + 1$$

Demonstration

Let Alice's EOA have a current account nonce N = 5. She signs Tx-A: transfer 1 ETH to Bob with nonce = 5.

Step	Event	N (state)	Outcome
1	Alice broadcasts Tx-A (nonce 5)	5	Nonce matches, so Tx-A is valid and included in a block.
2	Execution completes	6	Alice's account nonce becomes 6.
3	Attacker re-broadcasts the same signed Tx-A (nonce 5)	6	5 < 6, so it is rejected as 'nonce too low'. 1 ETH is not paid twice.
4	Alice sends Tx-B (nonce 6)	6 → 7	Valid; processed normally.
5	A transaction with nonce 8 arrives while N = 7	7	Gap: it waits in the pool until nonce 7 is processed.

Why the Attacker Cannot Bypass It

- **Nonce is signed:** altering the nonce in a copied transaction changes its hash, so the original signature no longer verifies for Alice's address and the transaction is invalid.
- **Uniqueness and order:** each nonce value can be used only once, so a transaction can be included at most once, and transactions are executed in the order of their nonces.
- **Replacement, not duplication:** a pending transaction can only be replaced by another with the same nonce and a higher gas price (transaction bumping); only one of them is ever executed.
- **Cross-chain replay:** since chains sharing a transaction format could accept the same signed transaction, Ethereum also includes the **chain ID** (EIP-155) in the signed payload, which complements the nonce.

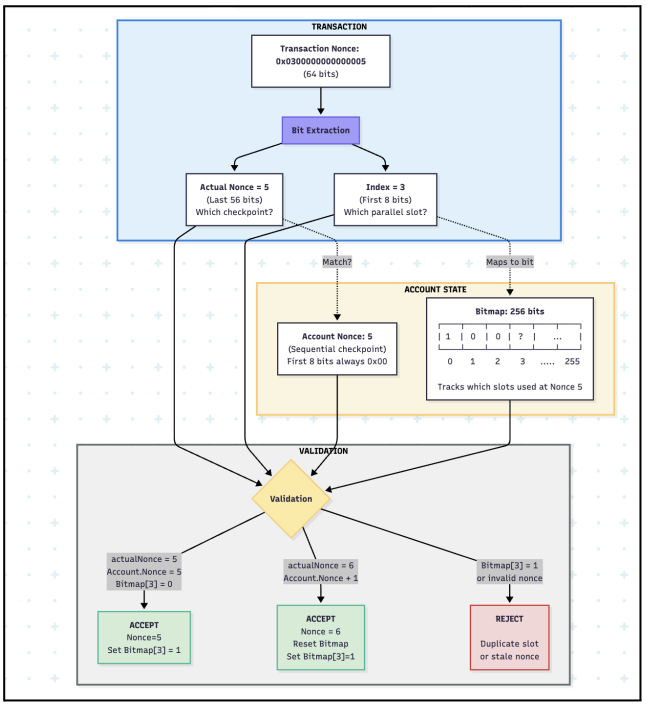


Fig. 1: Nonce-based replay protection in an Ethereum EOA

Conclusion

The nonce acts as a strictly increasing, signed sequence number. A valid transaction must carry exactly the account's current nonce, and the nonce increases after execution, so a previously processed transaction can never match again. This guarantees uniqueness and ordering of transactions from an EOA and prevents double execution of the same transaction.

M6/6

Q2. What are stablecoins? Explain their importance in the cryptocurrency and DeFi ecosystems.

A **stablecoin** is a cryptocurrency designed to keep a stable market value by being pegged to a reference asset, most commonly the US dollar (1 token ≈ 1 USD). Unlike Bitcoin or Ether, whose prices fluctuate sharply, stablecoins combine the speed and programmability of blockchains with the price stability of traditional money. Well-known examples are **USDT (Tether), USDC and DAI**.

Types of Stablecoins

Type	Stability Mechanism	Examples / Remarks
Fiat-collateralised	Backed 1:1 by reserves of fiat currency or cash equivalents held off-chain by an issuer; redeemable for fiat.	USDT, USDC. Centralised; relies on trust in the issuer's reserves.
Crypto-collateralised	Over-collateralised with crypto locked in smart contracts; liquidated if collateral value falls.	DAI (MakerDAO). Decentralised and transparent on-chain.
Algorithmic	No collateral; supply is expanded or contracted by an algorithm based on demand.	TerraUSD (UST) collapsed in 2022; regarded as high risk.
Commodity-backed	Backed by assets such as gold.	Hedge against inflation; smaller market share.

Key Formula

Collateral Ratio (%) = (Value of Collateral Locked ÷ Value of Stablecoin Issued) × 100

Example: for a minimum ratio of 150%, a user must lock ETH worth $150 to mint 100 DAI. The excess collateral absorbs price falls in ETH and keeps the peg safe. Fiat-backed coins aim at a ratio of at least 100% (reserves ≥ coins in circulation).

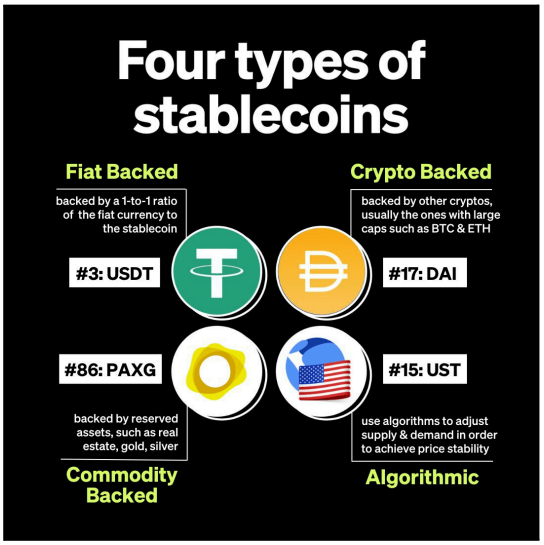


Fig. 2: Types of stablecoins and working of a crypto-collateralised stablecoin (DAI)

Importance in the Cryptocurrency Ecosystem

- **Volatility shield:** traders and investors can park value in a stable asset without cashing out to a bank.
- **Trading base pairs:** most exchange pairs are quoted in USDT or USDC, giving high liquidity.
- **Payments and remittances:** fast, low-cost cross-border transfers, available 24×7.
- **Bridge to traditional finance:** links fiat money with the crypto economy.

Importance in the DeFi Ecosystem

- **Collateral and lending:** stablecoins are borrowed and lent on protocols such as Aave and Compound, with predictable loan values.
- **Liquidity pools and DEXs:** stable pairs reduce impermanent loss and enable efficient swaps.
- **Yield generation:** holders earn interest or rewards by supplying stablecoins to protocols.
- **Unit of account for smart contracts:** loans, payroll and derivatives can be denominated in a stable value.

Challenges

Stablecoins carry risks of de-pegging (as in the UST collapse), lack of reserve transparency in centralised issuers, smart-contract vulnerabilities and regulatory uncertainty.

Conclusion

Stablecoins provide the price stability that cryptocurrencies lack, acting as the main medium of exchange, store of value and collateral in the blockchain economy. They are the backbone of DeFi liquidity and a key link between traditional finance and decentralised systems.

M6/26

Q3. Compare Web 2.0 and Web 3.0 based on their features and applications. Explain how Web 3.0 supports decentralized services with suitable examples.

Introduction

Web 2.0 (c. 2004 onwards) is the interactive, user-generated-content web run by large platforms. **Web 3.0** is the emerging decentralised web built on blockchain, smart contracts and peer-to-peer networks, in which users own their data and assets. Web 2.0 is often summarised as *Read/Write* and Web 3.0 as *Read/Write/Own*.

Comparison of Web 2.0 and Web 3.0

Feature	Web 2.0	Web 3.0
Control	Centralised; owned by companies	Decentralised; community and DAO governance
Architecture	Client–server applications	Decentralised apps (dApps) on blockchain nodes
Data storage	Company servers and cloud (AWS, Google Cloud)	Distributed storage: IPFS, Filecoin, Arweave
Data ownership	Platform owns user data	User owns data and assets via wallet keys
Identity	Email/password, social logins	Wallet-based, self-sovereign identity
Trust	Trusted intermediaries (banks, platforms)	Trustless; smart contracts enforce rules
Key technologies	HTML5, CSS3, AJAX, APIs, JS frameworks	Blockchain, smart contracts, cryptocurrency, AI/ML, IPFS
Monetisation	Advertising using user data	Tokens, NFTs, direct peer-to-peer payments
Weaknesses	Single point of failure, censorship, privacy risks	Scalability, complexity, regulatory uncertainty
Applications	Facebook, YouTube, Instagram, Gmail, Amazon	Uniswap, Aave, OpenSea, MetaMask, Filecoin

How Web 3.0 Supports Decentralised Services

A Web 3.0 application typically has a front end that talks to smart contracts on a blockchain such as Ethereum, with files kept on distributed storage and the user logging in with a crypto wallet. No single company runs the back end, so services stay available and rules cannot be changed silently. The supporting building blocks are:

- **Blockchain and consensus:** an immutable, shared ledger maintained by many nodes.
- **Smart contracts:** self-executing code that automates agreements without intermediaries.
- **Decentralised storage:** data is spread across many computers (IPFS, Filecoin).
- **Wallets and tokens:** the user's key is their identity and payment method.

Examples of Decentralised Services

- **DeFi:** Uniswap (token exchange) and Aave (lending and borrowing) run through smart contracts without a bank or broker.
- **NFT marketplaces:** OpenSea lets artists mint NFTs, with royalties paid automatically by smart contracts.

- **Decentralised storage:** IPFS and Filecoin replace centralised cloud storage.
- **Decentralised social media:** Lens Protocol keeps profiles and content under user ownership.
- **Wallets and identity:** MetaMask gives access to dApps without creating accounts.

Conclusion

Web 2.0 offers convenience and rich interaction but concentrates data and power in a few companies. Web 3.0 shifts control to users through blockchain, smart contracts and decentralised storage, enabling trustless services such as DeFi, NFTs and decentralised storage, though scalability, usability and regulation remain open challenges.